

Appendix

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This appendix gives the complete derivation of the fixed and linear-parameter-varying equivalent-circuit models used in this study [1, 6, 7]. It establishes the state recursion, state-of-charge feasibility conditions, physical parameter constraints, hierarchical identifiability, candidate-family dimensions, sparse-support closure, calibration-stage selection rule, and test-isolation properties. The trajectory and chemistry structure follows the open randomized-current lithium-ion degradation dataset used in the main study [?].

Complete trajectories are indexed by $s \in \{1, \dots, S\}$. Trajectory s contains N_s ordered samples indexed by

$$k = 0, 1, \dots, N_s - 1.$$

The elapsed time, measured terminal voltage, current, temperature, and reconstructed state of charge are denoted by

$$t_{s,k}, \quad V_{s,k}, \quad I_{s,k}, \quad T_{s,k}, \quad z_{s,k},$$

respectively. The sampling interval is

$$\Delta t_{s,k} = t_{s,k+1} - t_{s,k}, \quad \Delta t_{s,k} > 0. \quad (\text{A.1})$$

Positive current denotes discharge:

$$I_{s,k} > 0 \iff \text{discharge}. \quad (\text{A.2})$$

A.1. State-of-charge reconstruction

The signed charge transferred over the interval $[t_{s,k}, t_{s,k+1}]$ is evaluated by trapezoidal integration:

$$\Delta q_{s,k} = \frac{\Delta t_{s,k} (I_{s,k} + I_{s,k+1})}{2 \times 3600}, \quad k = 0, \dots, N_s - 2, \quad (\text{A.3})$$

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where $\Delta q_{s,k}$ is expressed in ampere-hours. The cumulative signed charge is

$$q_{s,0}^{\text{cum}} = 0, \quad q_{s,k}^{\text{cum}} = \sum_{n=0}^{k-1} \Delta q_{s,n}, \quad k \geq 1. \quad (\text{A.4})$$

Let $Q_s > 0$ denote the capacity assigned to trajectory s , and let η_{cc} denote the Coulombic-efficiency factor. The implemented analysis uses

$$\eta_{\text{cc}} = 1. \quad (\text{A.5})$$

Define the cumulative SOC displacement as

$$d_{s,k} = \frac{\eta_{\text{cc}} q_{s,k}^{\text{cum}}}{Q_s}. \quad (\text{A.6})$$

The reconstructed SOC is then

$$z_{s,k} = z_{s,0}^{\text{sel}} - d_{s,k}, \quad (\text{A.7})$$

or, equivalently,

$$z_{s,k+1} = z_{s,k} - \frac{\eta_{\text{cc}} \Delta t_{s,k} (I_{s,k} + I_{s,k+1})}{7200 Q_s}. \quad (\text{A.8})$$

Equation (A.8) is consistent with the adopted current convention. During discharge, $I_{s,k} > 0$, so the reconstructed SOC decreases. During charge, $I_{s,k} < 0$, so the reconstructed SOC increases.

A.2. Charge-consistent initial-SOC interval

For chemistry c_s , let the validated OCV support be

$$z_{\min}^{(c_s)} \leq z \leq z_{\max}^{(c_s)}. \quad (\text{A.9})$$

Because

$$z_{s,k} = z_{s,0}^{\text{sel}} - d_{s,k},$$

requiring every reconstructed SOC value to remain within the validated support gives

$$z_{\min}^{(c_s)} \leq z_{s,0}^{\text{sel}} - d_{s,k} \leq z_{\max}^{(c_s)} \quad \text{for all } k. \quad (\text{A.10})$$

Rearranging the two inequalities yields

$$z_{\min}^{(c_s)} + d_{s,k} \leq z_{s,0}^{\text{sel}} \leq z_{\max}^{(c_s)} + d_{s,k}. \quad (\text{A.11})$$

The initial SOC must therefore belong to

$$\mathcal{Z}_{s,0}^{\text{feas}} = \bigcap_{k=0}^{N_s-1} \left[z_{\min}^{(c_s)} + d_{s,k}, z_{\max}^{(c_s)} + d_{s,k} \right]. \quad (\text{A.12})$$

The intersection is

$$\mathcal{Z}_{s,0}^{\text{feas}} = \left[z_{\min}^{(c_s)} + \max_k d_{s,k}, z_{\max}^{(c_s)} + \min_k d_{s,k} \right]. \quad (\text{A.13})$$

Proof.. For each k , the lower bound on $z_{s,0}^{\text{sel}}$ is $z_{\min}^{(c_s)} + d_{s,k}$. Satisfying all lower bounds is equivalent to satisfying their maximum. Similarly, satisfying all upper bounds is equivalent to remaining below their minimum. Hence, Eq. (A.13) follows. The interval is nonempty if and only if

$$z_{\min}^{(c_s)} + \max_k d_{s,k} \leq z_{\max}^{(c_s)} + \min_k d_{s,k}. \quad (\text{A.14})$$

□

A.3. OCV-derived initial-SOC interval

Chemistry-specific OCV construction and inversion follow the established role of OCV–SOC characterization in equivalent-circuit modelling [2, 3]. Let the chemistry-specific OCV relation be represented by

$$\left\{ \left(z_j^{(c)}, U_j^{(c)} \right) \right\}_{j=1}^{M_c}. \quad (\text{A.15})$$

The forward OCV map is

$$U_{\text{oc}}^{(c)}(z) = \text{interp} \left(z; \mathbf{z}_{\text{grid}}^{(c)}, \mathbf{U}_{\text{grid}}^{(c)} \right). \quad (\text{A.16})$$

Before forward interpolation, SOC is restricted to the validated chemistry-specific support:

$$z_{s,k}^{\text{OCV}} = \text{clip} \left(z_{s,k}; z_{\min}^{(c_s)}, z_{\max}^{(c_s)} \right), \quad (\text{A.17})$$

where

$$\text{clip}(x; a, b) = \min(\max(x, a), b). \quad (\text{A.18})$$

The OCV input to the ECM is

$$U_{\text{oc},s,k} = U_{\text{oc}}^{(c_s)} \left(z_{s,k}^{\text{OCV}} \right). \quad (\text{A.19})$$

Nearly flat OCV regions are treated as plateau-equivalent. The voltage tolerance is

$$\epsilon_{\text{OCV}} = \max(10^{-8}, 10\epsilon_{\text{mon}}), \quad \epsilon_{\text{mon}} = 10^{-9} \text{ V}. \quad (\text{A.20})$$

For a rest-voltage estimate $V_{s,0}^{\text{rest}}$, define the set-valued inverse

$$\mathcal{Z}_{s,0}^{\text{OCV}} = \left\{ z \in [z_{\min}^{(c_s)}, z_{\max}^{(c_s)}] : |U_{\text{oc}}^{(c_s)}(z) - V_{s,0}^{\text{rest}}| \leq \epsilon_{\text{OCV}} \right\}. \quad (\text{A.21})$$

For a validated monotone OCV relation, this set is either empty, a single point, or a closed SOC interval:

$$\mathcal{Z}_{s,0}^{\text{OCV}} = [z_{s,0}^{\text{OCV,lo}}, z_{s,0}^{\text{OCV,hi}}]. \quad (\text{A.22})$$

The OCV-derived and charge-consistent evidence are combined through

$$\mathcal{Z}_{s,0}^{\text{con}} = \mathcal{Z}_{s,0}^{\text{OCV}} \cap \mathcal{Z}_{s,0}^{\text{feas}}. \quad (\text{A.23})$$

If this intersection is nonempty, a representative OCV estimate $z_{s,0}^{\text{OCV,rep}}$ is projected onto the consistent interval:

$$z_{s,0}^{\text{sel}} = \text{clip}\left(z_{s,0}^{\text{OCV,rep}}, z_{s,0}^{\text{con,lo}}, z_{s,0}^{\text{con,hi}}\right). \quad (\text{A.24})$$

If the intersection is empty, the trajectory is not admitted as parameter-fitting evidence.

The OCV relation, capacity assignment, and reconstructed SOC are fixed before candidate fitting. They are not free parameters of the ECM or LPV optimizer.

A.4. Continuous-time RC branch and exact discrete recursion

For polarization branch j , the continuous-time equation is

$$C_{j,s,k} \frac{dv_j(t)}{dt} = I_{s,k} - \frac{v_j(t)}{R_{j,s,k}}, \quad t \in [t_{s,k}, t_{s,k+1}), \quad (\text{A.25})$$

where the current and parameter values evaluated at sample k are held constant over the interval. Define

$$\tau_{j,s,k} = R_{j,s,k} C_{j,s,k}. \quad (\text{A.26})$$

Equation (A.25) becomes

$$\frac{dv_j(t)}{dt} + \frac{1}{\tau_{j,s,k}} v_j(t) = \frac{R_{j,s,k}}{\tau_{j,s,k}} I_{s,k}. \quad (\text{A.27})$$

Applying the integrating factor $\exp(t/\tau_{j,s,k})$ gives

$$\begin{aligned} v_{j,s,k+1} &= \exp\left(-\frac{\Delta t_{s,k}}{\tau_{j,s,k}}\right) v_{j,s,k} \\ &+ R_{j,s,k} \left[1 - \exp\left(-\frac{\Delta t_{s,k}}{\tau_{j,s,k}}\right)\right] I_{s,k}. \end{aligned} \quad (\text{A.28})$$

Define

$$\alpha_{j,s,k} = \exp\left(-\frac{\Delta t_{s,k}}{\tau_{j,s,k}}\right). \quad (\text{A.29})$$

The recursive update is

$$v_{j,s,k+1} = \alpha_{j,s,k} v_{j,s,k} + R_{j,s,k} (1 - \alpha_{j,s,k}) I_{s,k}. \quad (\text{A.30})$$

For numerical accuracy when $\Delta t_{s,k}/\tau_{j,s,k}$ is small,

$$1 - \alpha_{j,s,k} = -\expm1\left(-\frac{\Delta t_{s,k}}{\tau_{j,s,k}}\right). \quad (\text{A.31})$$

A.5. Fixed 1RC and 2RC voltage models

The following 1RC and 2RC structures are standard low-order Thévenin representations used in lithium-ion battery modelling [1?, 4, 5]. For a J -branch equivalent-circuit model,

$$\widehat{V}_{s,k}^{(J)} = U_{\text{oc},s,k} - R_{0,s,k} I_{s,k} - \sum_{j=1}^J v_{j,s,k}. \quad (\text{A.32})$$

For the fixed 1RC model,

$$\widehat{V}_{s,k}^{\text{1RC}} = U_{\text{oc},s,k} - R_0 I_{s,k} - v_{1,s,k}, \quad (\text{A.33})$$

with

$$v_{1,s,k+1} = \alpha_{1,s,k} v_{1,s,k} + R_1 (1 - \alpha_{1,s,k}) I_{s,k}. \quad (\text{A.34})$$

For the fixed 2RC model,

$$\widehat{V}_{s,k}^{\text{2RC}} = U_{\text{oc},s,k} - R_0 I_{s,k} - v_{1,s,k} - v_{2,s,k}, \quad (\text{A.35})$$

with Eq. (A.30) applied for $j \in \{1, 2\}$.

A.6. Stability and boundedness of the RC recursion

Because

$$\begin{aligned} \Delta t_{s,k} &> 0 \quad \text{and} \quad \tau_{j,s,k} > 0, \\ 0 &< \alpha_{j,s,k} < 1. \end{aligned} \tag{A.36}$$

Proof. For any $x > 0$, $0 < \exp(-x) < 1$. Taking $x = \Delta t_{s,k}/\tau_{j,s,k} > 0$ gives Eq. (A.36). $\square$

Suppose that

$$|I_{s,k}| \leq I_{\max}, \quad 0 < R_{j,s,k} \leq R_{\max}.$$

Then

$$|v_{j,s,k+1}| \leq \alpha_{j,s,k} |v_{j,s,k}| + (1 - \alpha_{j,s,k}) R_{\max} I_{\max}. \tag{A.37}$$

It follows inductively that

$$|v_{j,s,k}| \leq \max(|v_{j,s,0}|, R_{\max} I_{\max}). \tag{A.38}$$

Thus, bounded current and bounded positive resistance imply bounded polarization states.

If

$$\Delta t_{s,k} \geq \Delta t_{\min} > 0, \quad \tau_{j,s,k} \leq \tau_{\max} < \infty,$$

then

$$\alpha_{j,s,k} \leq \exp\left(-\frac{\Delta t_{\min}}{\tau_{\max}}\right) = \bar{\alpha} < 1, \tag{A.39}$$

which gives uniform exponential contraction of the homogeneous state recursion.

A.7. Polarization-state initialization

A rest start is detected when

$$|I_{s,k}| \leq I_{\text{rest}}, \quad k = 0, \dots, N_{\text{rest}} - 1, \tag{A.40}$$

with

$$I_{\text{rest}} = 0.05 \text{ A}, \quad N_{\text{rest}} = 5. \tag{A.41}$$

For a detected rest start,

$$v_{1,s,0} = v_{2,s,0} = 0. \tag{A.42}$$

Otherwise, the total initial polarization voltage is

$$v_{p,s,0}^{\text{tot}} = U_{\text{oc},s,0} - R_{0,s,0}I_{s,0} - V_{s,0}. \quad (\text{A.43})$$

The deterministic resistance-weighted allocation is

$$v_{1,s,0} = \frac{R_{1,s,0}}{R_{1,s,0} + R_{2,s,0}} v_{p,s,0}^{\text{tot}}, \quad (\text{A.44})$$

and

$$v_{2,s,0} = \frac{R_{2,s,0}}{R_{1,s,0} + R_{2,s,0}} v_{p,s,0}^{\text{tot}}. \quad (\text{A.45})$$

Since $R_{1,s,0} > 0$ and $R_{2,s,0} > 0$,

$$v_{1,s,0} + v_{2,s,0} = v_{p,s,0}^{\text{tot}}. \quad (\text{A.46})$$

Therefore,

$$\begin{aligned} \widehat{V}_{s,0} &= U_{\text{oc},s,0} - R_{0,s,0}I_{s,0} - v_{p,s,0}^{\text{tot}} \\ &= V_{s,0}. \end{aligned} \quad (\text{A.47})$$

The decomposition between the two RC branches is not uniquely identifiable from a single voltage observation. The resistance-weighted allocation is therefore a reproducible initialization policy, not a proof of the true physical branch states. The first sample is excluded from the fitting loss.

A.8. Training-only SOC-temperature scheduler

State-dependent and LPV battery formulations motivate scheduling the electrical representation with measured or reconstructed operating variables [6?, 8, 9, 7]. The implemented LPV scheduling vector contains reconstructed SOC and measured temperature only:

$$\mathbf{q}_{s,k} = \begin{bmatrix} z_{s,k} \\ T_{s,k} \end{bmatrix}. \quad (\text{A.48})$$

Signed current, current magnitude, current direction, and elapsed time are diagnostic variables rather than continuous scheduling coordinates. Chemistry and cell identity enter the hierarchical coefficient representation.

Let $\mathbf{q}_{\min}^{\text{tr}}$ and $\mathbf{q}_{\max}^{\text{tr}}$ denote the componentwise limits calculated from the training partition. Define

$$\mathbf{q}_{\text{mid}} = \frac{\mathbf{q}_{\max}^{\text{tr}} + \mathbf{q}_{\min}^{\text{tr}}}{2}, \quad (\text{A.49})$$

and

$$\mathbf{q}_{\text{half}} = \frac{\mathbf{q}_{\text{max}}^{\text{tr}} - \mathbf{q}_{\text{min}}^{\text{tr}}}{2}. \quad (\text{A.50})$$

Each scheduler coordinate must satisfy

$$q_{\text{half},i} > 0. \quad (\text{A.51})$$

The normalized scheduler is

$$\boldsymbol{\xi}_{s,k} = (\mathbf{q}_{s,k} - \mathbf{q}_{\text{mid}}) \oslash \mathbf{q}_{\text{half}} = \begin{bmatrix} \xi_{z,s,k} \\ \xi_{T,s,k} \end{bmatrix}, \quad (\text{A.52})$$

where $\oslash$ denotes componentwise division.

For the implemented training partition,

$$z_{\text{min}}^{\text{tr}} = 0, \quad z_{\text{max}}^{\text{tr}} = 1, \quad (\text{A.53})$$

and

$$T_{\text{min}}^{\text{tr}} = 23.5929^\circ\text{C}, \quad T_{\text{max}}^{\text{tr}} = 57.9206^\circ\text{C}. \quad (\text{A.54})$$

Thus,

$$\xi_{z,s,k} = \frac{z_{s,k} - 0.5}{0.5}, \quad (\text{A.55})$$

and

$$\xi_{T,s,k} = \frac{T_{s,k} - 40.75675}{17.16385}. \quad (\text{A.56})$$

The normalized scheduler is not clipped to $[-1, 1]^2$. Held-out samples outside the training range therefore remain outside the normalized training domain.

The frozen basis is

$$\phi(\boldsymbol{\xi}_{s,k}) = \begin{bmatrix} 1 \\ \xi_{z,s,k} \\ \xi_{T,s,k} \\ \xi_{z,s,k}^2 \\ \xi_{T,s,k}^2 \\ \xi_{z,s,k} \xi_{T,s,k} \end{bmatrix} \in \mathbb{R}^6. \quad (\text{A.57})$$

A.9. Transformed physical parameters

The transformed representation is consistent with battery ECM studies in which resistance and polarization parameters vary with SOC, temperature, chemistry, and operating condition [4?, 5]. The transformed channels are

$$\boldsymbol{\eta}_{s,k} = \begin{bmatrix} \eta_{R_0,s,k} \\ \eta_{R_1,s,k} \\ \eta_{R_2,s,k} \\ \eta_{\tau_1,s,k} \\ \eta_{\Delta\tau,s,k} \end{bmatrix}. \quad (\text{A.58})$$

For channel r ,

$$\tilde{\eta}_{r,s,k} = \text{clip}(\eta_{r,s,k}; \eta_r^{\min}, \eta_r^{\max}). \quad (\text{A.59})$$

The physical parameters used by the simulator are

$$R_{0,s,k} = \exp(\tilde{\eta}_{R_0,s,k}), \quad (\text{A.60})$$

$$R_{1,s,k} = \exp(\tilde{\eta}_{R_1,s,k}), \quad R_{2,s,k} = \exp(\tilde{\eta}_{R_2,s,k}), \quad (\text{A.61})$$

$$\tau_{1,s,k} = \exp(\tilde{\eta}_{\tau_1,s,k}), \quad (\text{A.62})$$

$$\Delta\tau_{s,k} = \exp(\tilde{\eta}_{\Delta\tau,s,k}), \quad (\text{A.63})$$

and

$$\tau_{2,s,k} = \tau_{1,s,k} + \Delta\tau_{s,k}. \quad (\text{A.64})$$

The frozen physical intervals are

$$10^{-5} \leq R_{0,s,k}, R_{1,s,k}, R_{2,s,k} \leq 1 \quad [\Omega], \quad (\text{A.65})$$

$$10^{-2} \leq \tau_{1,s,k} \leq 2 \times 10^3 \quad [\text{s}], \quad (\text{A.66})$$

and

$$10^{-2} \leq \Delta\tau_{s,k} \leq 2 \times 10^4 \quad [\text{s}]. \quad (\text{A.67})$$

Positivity and ordering proof. For any real x , $\exp(x) > 0$. Hence,

$$R_{0,s,k} > 0, \quad R_{1,s,k} > 0, \quad R_{2,s,k} > 0, \quad \tau_{1,s,k} > 0, \quad \Delta\tau_{s,k} > 0.$$

Since

$$\begin{aligned} \tau_{2,s,k} &= \tau_{1,s,k} + \Delta\tau_{s,k}, \\ 0 &< \tau_{1,s,k} < \tau_{2,s,k}. \end{aligned} \quad (\text{A.68})$$

The transformation therefore enforces positive physical parameters and removes branch-label exchange between unordered time constants. $\square$

The simulator uses the clipped predictors $\tilde{\eta}_{r,s,k}$, while the objective penalizes excursions of the unclipped predictors $\eta_{r,s,k}$.

A.10. Hierarchical SOC-temperature parameter map

Let

$$\mathcal{R} = \{R_0, R_1, R_2, \tau_1, \Delta\tau\}.$$

For chemistry c_s , cell ℓ_s , and basis term m ,

$$\eta_{r,s,k} = \sum_{m=1}^6 (\beta_{r,m}^G + \beta_{r,c_s,m}^C + \beta_{r,c_s,\ell_s,m}^L) \phi_m(\boldsymbol{\xi}_{s,k}). \quad (\text{A.69})$$

Here, $\beta_{r,m}^G$ is the global coefficient, $\beta_{r,c,m}^C$ is the chemistry-level deviation, and $\beta_{r,c,\ell,m}^L$ is the cell-level deviation nested within chemistry c .

Let $n_{c,\ell} > 0$ denote the training representation count for cell ℓ in chemistry c , and define

$$N_c = \sum_{\ell \in \mathcal{L}_c} n_{c,\ell}, \quad N = \sum_{c=1}^C N_c. \quad (\text{A.70})$$

Weighted centring removes the nonuniqueness between global and nested deviation terms. Related structured and hierarchical sparsity principles are discussed in [? 7]. For each transformed channel and basis term, impose

$$\sum_{c=1}^C N_c \beta_{r,c,m}^C = 0, \quad (\text{A.71})$$

and

$$\sum_{\ell \in \mathcal{L}_c} n_{c,\ell} \beta_{r,c,\ell,m}^L = 0. \quad (\text{A.72})$$

A.11. Uniqueness of the hierarchical decomposition

For fixed transformed channel and basis term, suppress r, m and define the effective cell coefficient as

$$\theta_{c,\ell} = \beta^G + \beta_c^C + \beta_{c,\ell}^L. \quad (\text{A.73})$$

Proposition A.1.. Under positive training weights and the weighted zero-sum constraints, the global, chemistry-level, and cell-level decomposition is unique.

Proof. The weighted grand mean is

$$\begin{aligned}\bar{\theta} &= \frac{1}{N} \sum_{c=1}^C \sum_{\ell \in \mathcal{L}_c} n_{c,\ell} \theta_{c,\ell} \\ &= \beta^G + \frac{1}{N} \sum_{c=1}^C N_c \beta_c^C + \frac{1}{N} \sum_{c=1}^C \sum_{\ell \in \mathcal{L}_c} n_{c,\ell} \beta_{c,\ell}^L.\end{aligned}\tag{A.74}$$

The final two terms vanish by Eqs. (A.71) and (A.72). Therefore,

$$\beta^G = \bar{\theta}.\tag{A.75}$$

Within chemistry c ,

$$\begin{aligned}\bar{\theta}_c &= \frac{1}{N_c} \sum_{\ell \in \mathcal{L}_c} n_{c,\ell} \theta_{c,\ell} \\ &= \beta^G + \beta_c^C + \frac{1}{N_c} \sum_{\ell \in \mathcal{L}_c} n_{c,\ell} \beta_{c,\ell}^L.\end{aligned}\tag{A.76}$$

The final term vanishes, so

$$\beta_c^C = \bar{\theta}_c - \beta^G.\tag{A.77}$$

Finally,

$$\beta_{c,\ell}^L = \theta_{c,\ell} - \beta^G - \beta_c^C = \theta_{c,\ell} - \bar{\theta}_c.\tag{A.78}$$

Every component is therefore uniquely determined. $\square$

A.12. Reduced identifiable coordinates

Let $\mathbf{w}_C \in \mathbb{R}^C$ contain the positive chemistry weights N_c . An orthonormal null-space matrix $\mathbf{N}_C \in \mathbb{R}^{C \times (C-1)}$ satisfies

$$\mathbf{w}_C^\top \mathbf{N}_C = \mathbf{0}^\top, \quad \mathbf{N}_C^\top \mathbf{N}_C = \mathbf{I}.\tag{A.79}$$

Every chemistry-deviation vector satisfying the weighted zero-sum constraint can be represented as

$$\boldsymbol{\beta}^C = \mathbf{N}_C \boldsymbol{\gamma}^C.\tag{A.80}$$

For chemistry c , let $\mathbf{w}_{L,c}$ contain the positive cell weights. An orthonormal null-space matrix satisfies

$$\mathbf{w}_{L,c}^\top \mathbf{N}_{L,c} = \mathbf{0}^\top.\tag{A.81}$$

The cell deviations are

$$\boldsymbol{\beta}_c^L = \mathbf{N}_{L,c} \boldsymbol{\gamma}_c^L. \quad (\text{A.82})$$

Collecting all transformed channels and basis terms gives

$$\boldsymbol{\beta}^{\text{full}} = \mathbf{A}_{\text{id}} \boldsymbol{\beta}^{\text{red}}, \quad (\text{A.83})$$

where $\mathbf{A}_{\text{id}}$ is fixed before fitting.

For one transformed channel and one basis term,

$$\begin{aligned} p_{\text{one}} &= 1 + (C - 1) + \sum_{c=1}^C (|\mathcal{L}_c| - 1) \\ &= 1 + (C - 1) + (L - C) \\ &= L. \end{aligned} \quad (\text{A.84})$$

With $C = 3$ chemistries and $L = 8$ cells,

$$p_{\text{one}} = 8. \quad (\text{A.85})$$

With five transformed channels and six basis terms,

$$p_{\text{red}} = 5 \times 6 \times 8 = 240. \quad (\text{A.86})$$

The nominal unconstrained full dimension is

$$p_{\text{full}} = 5 \times 6 \times (1 + 3 + 8) = 360. \quad (\text{A.87})$$

A.13. Candidate-family definitions

The candidate set is

$$\mathcal{M} = \{M_0, M_1, M_2, M_3, M_4\}. \quad (\text{A.88})$$

Each candidate is represented by a reduced-coordinate mask

$$\mathbf{m}_M^{\text{cand}} \in \{0, 1\}^{240}. \quad (\text{A.89})$$

The candidate-enabled vector is

$$\boldsymbol{\beta}_M = \mathbf{m}_M^{\text{cand}} \odot \boldsymbol{\beta}^{\text{red}}. \quad (\text{A.90})$$

A.13.1. Candidate M_0 : fixed 2RC model

Only the five global intercepts are enabled:

$$\eta_{r,s,k} = \beta_{r,1}^G, \quad r \in \mathcal{R}, \quad (\text{A.91})$$

so

$$p_{M_0} = 5. \quad (\text{A.92})$$

A.13.2. Candidate M_1 : global dense LPV model

All six global basis terms are enabled:

$$\eta_{r,s,k} = \sum_{m=1}^6 \beta_{r,m}^G \phi_m(\boldsymbol{\xi}_{s,k}), \quad (\text{A.93})$$

and

$$p_{M_1} = 5 \times 6 = 30. \quad (\text{A.94})$$

A.13.3. Candidate M_2 : global sparse LPV model

Candidate M_2 has the same enabled coefficient space as M_1 :

$$\mathcal{A}_{M_2} = \mathcal{A}_{M_1}, \quad p_{M_2} = 30, \quad (\text{A.95})$$

but applies sparse-group regularization to the nonconstant global basis terms. There are five nonconstant basis terms for each of five transformed channels:

$$|\mathcal{G}_{M_2}| = 5 \times 5 = 25. \quad (\text{A.96})$$

A.13.4. Candidate M_3 : hierarchical dense LPV model

Candidate M_3 enables the complete identifiable hierarchy:

$$\eta_{r,s,k} = \sum_{m=1}^6 (\beta_{r,m}^G + \beta_{r,c_s,m}^C + \beta_{r,c_s,\ell_s,m}^L) \phi_m(\boldsymbol{\xi}_{s,k}), \quad (\text{A.97})$$

with

$$p_{M_3}^{\text{full}} = 360, \quad p_{M_3}^{\text{red}} = 240. \quad (\text{A.98})$$

A.13.5. Candidate M_4 : hierarchical sparse LPV model

Candidate M_4 has the same enabled identifiable space as M_3 :

$$\mathcal{A}_{M_4} = \mathcal{A}_{M_3}, \quad p_{M_4}^{\text{red}} = 240. \quad (\text{A.99})$$

The sparse group set contains

25

nonconstant global groups,

30

chemistry-deviation groups, and

90

cell-deviation groups. Hence,

$$|\mathcal{G}_{M_4}| = 25 + 30 + 90 = 145. \quad (\text{A.100})$$

A.14. Nestedness of the candidate spaces

The coefficient spaces satisfy

$$\mathcal{A}_{M_0} \subset \mathcal{A}_{M_1} = \mathcal{A}_{M_2} \subset \mathcal{A}_{M_3} = \mathcal{A}_{M_4}. \quad (\text{A.101})$$

Proof. Candidate M_0 is obtained from M_1 by setting all nonconstant global coefficients to zero. Candidates M_1 and M_2 enable the same global space and differ only in regularization. Candidate M_1 is obtained from M_3 by setting all chemistry- and cell-level deviations to zero. Candidates M_3 and M_4 enable the same hierarchical space and differ only in regularization. $\square$

A.15. Initialization from the fixed 2RC benchmark

Every LPV candidate is initialized from the fixed 2RC parameters:

$$\eta_{R_i}^{(0)} = \log R_i^{\text{fixed}}, \quad i \in \{0, 1, 2\}, \quad (\text{A.102})$$

$$\eta_{\tau_1}^{(0)} = \log \tau_1^{\text{fixed}}, \quad (\text{A.103})$$

and

$$\eta_{\Delta\tau}^{(0)} = \log (\tau_2^{\text{fixed}} - \tau_1^{\text{fixed}}). \quad (\text{A.104})$$

These values initialize the global intercepts. All nonconstant, chemistry-level, and cell-level deviations begin at zero.

A.16. Residual and robust data loss

The signed residual is

$$e_{s,k} = V_{s,k} - \widehat{V}_{s,k}. \quad (\text{A.105})$$

Thus, $e_{s,k} < 0$ denotes voltage overprediction and $e_{s,k} > 0$ denotes voltage underprediction.

Let

$$m_{s,k}^{\text{loss}} \in \{0, 1\} \quad (\text{A.106})$$

denote the valid loss mask, with

$$m_{s,0}^{\text{loss}} = 0. \quad (\text{A.107})$$

The robust loss follows the Huber robust-estimation principle [10]. The pseudo-Huber loss is

$$\ell_{\text{PH}}(e; \delta) = \delta^2 \left[\sqrt{1 + \left(\frac{e}{\delta}\right)^2} - 1 \right], \quad \delta = 0.020 \text{ V}. \quad (\text{A.108})$$

Its first derivative is

$$\frac{\partial \ell_{\text{PH}}}{\partial e} = \frac{e}{\sqrt{1 + (e/\delta)^2}}, \quad (\text{A.109})$$

and its second derivative is

$$\frac{\partial^2 \ell_{\text{PH}}}{\partial e^2} = \frac{1}{[1 + (e/\delta)^2]^{3/2}} > 0. \quad (\text{A.110})$$

The scalar loss is smooth and strictly convex in e , although the complete recursive model remains nonlinear and nonconvex in its coefficient vector.

Let

$$N_{\text{loss}} = \sum_{s,k} m_{s,k}^{\text{loss}}. \quad (\text{A.111})$$

The normalized data term is

$$\mathcal{L}_{\text{data}} = \frac{\sum_{s,k} m_{s,k}^{\text{loss}} \ell_{\text{PH}}(e_{s,k}; \delta)}{\max(N_{\text{loss}}, 1)}. \quad (\text{A.112})$$

A.17. Ridge, sparse-group, and physical-range penalties

The ridge term is

$$\mathcal{L}_{\text{ridge},M} = \lambda_{\text{ridge}} \frac{\|\mathbf{m}_M^{\text{cand}} \odot \boldsymbol{\beta}^{\text{red}}\|_2^2}{240}, \quad \lambda_{\text{ridge}} = 10^{-7}. \quad (\text{A.113})$$

Grouped coefficient selection follows the group-lasso principle [?]. For $M \in \{M_2, M_4\}$, the sparse-group penalty is

$$\mathcal{P}_M(\boldsymbol{\beta}) = \sum_{g \in \mathcal{G}_M} \sqrt{|g|} \|\boldsymbol{\beta}_g\|_2. \quad (\text{A.114})$$

For a nonzero group,

$$\nabla_{\boldsymbol{\beta}_g} \left[\sqrt{|g|} \|\boldsymbol{\beta}_g\|_2 \right] = \sqrt{|g|} \frac{\boldsymbol{\beta}_g}{\|\boldsymbol{\beta}_g\|_2}. \quad (\text{A.115})$$

At $\boldsymbol{\beta}_g = \mathbf{0}$, the subdifferential is

$$\partial \left[\sqrt{|g|} \|\boldsymbol{\beta}_g\|_2 \right] = \left\{ \mathbf{u} : \|\mathbf{u}\|_2 \leq \sqrt{|g|} \right\}. \quad (\text{A.116})$$

The implementation uses the zero subgradient.

Define

$$d_{s,k,r}^- = \max(\eta_r^{\min} - \eta_{r,s,k}, 0), \quad (\text{A.117})$$

and

$$d_{s,k,r}^+ = \max(\eta_{r,s,k} - \eta_r^{\max}, 0). \quad (\text{A.118})$$

The physical-range penalty is

$$\mathcal{B}_{\text{phys}} = \frac{\sum_{s,k} m_{s,k}^{\text{loss}} \sum_{r \in \mathcal{R}} \left[(d_{s,k,r}^-)^2 + (d_{s,k,r}^+)^2 \right]}{5 \max(N_{\text{loss}}, 1)}. \quad (\text{A.119})$$

The complete objective is

$$\mathcal{J}_M = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{ridge},M} + \lambda_M \mathcal{P}_M + \lambda_{\text{phys}} \mathcal{B}_{\text{phys}}, \quad (\text{A.120})$$

with

$$\lambda_{\text{phys}} = 10^{-3}. \quad (\text{A.121})$$

For dense candidates,

$$\lambda_M = 0, \quad M \in \{M_0, M_1, M_3\}, \quad (\text{A.122})$$

whereas sparse candidates use

$$\lambda_M \in \{10^{-6}, 3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}, 10^{-4}\}. \quad (\text{A.123})$$

A.18. Optimization configuration

The candidate jobs are optimized with Adam [?]. Adam uses

$$\alpha_{\text{Adam}} = 2 \times 10^{-3}, \quad \beta_1 = 0.9, \quad \beta_2 = 0.999, \quad \epsilon_{\text{Adam}} = 10^{-8}. \quad (\text{A.124})$$

The trajectory-batch size is

$$B_{\text{traj}} = 8. \quad (\text{A.125})$$

For raw gradient $\mathbf{g}_t$,

$$\tilde{\mathbf{g}}_t = \mathbf{g}_t \min \left(1, \frac{10}{\|\mathbf{g}_t\|_2} \right), \quad (\text{A.126})$$

so

$$\|\tilde{\mathbf{g}}_t\|_2 \leq 10. \quad (\text{A.127})$$

Dense jobs use 1800 update steps, sparse jobs use 2400 update steps, and the selected-support refit uses 1200 update steps.

A.19. Support thresholding and heredity closure

A scalar coefficient is active when

$$|\beta_i| > \epsilon_{\text{support}}, \quad (\text{A.128})$$

and a sparse group is active when

$$\|\boldsymbol{\beta}_g\|_2 > \epsilon_{\text{support}}, \quad (\text{A.129})$$

where

$$\epsilon_{\text{support}} = 10^{-4}. \quad (\text{A.130})$$

The closure rule applies a strong-heredity interpretation to polynomial main effects and interactions [?]. Basis heredity requires

$$\xi_z^2 \implies \xi_z, \quad (\text{A.131})$$

$$\xi_T^2 \implies \xi_T, \quad (\text{A.132})$$

and

$$\xi_z \xi_T \implies \{\xi_z, \xi_T\}. \quad (\text{A.133})$$

Hierarchy heredity requires

$$\text{CELL} \implies \text{CHEMISTRY} \implies \text{GLOBAL}. \quad (\text{A.134})$$

Let $\mathcal{A}_{\text{raw}}$ denote the thresholded support and let

$$\text{cl}(\mathcal{A}_{\text{raw}}) \quad (\text{A.135})$$

denote the smallest support containing $\mathcal{A}_{\text{raw}}$ and satisfying all heredity rules.

Proposition A.2.. The heredity-closure operator is extensive, monotone, idempotent, and minimal.

Proof.. It is extensive because

$$\mathcal{A} \subseteq \text{cl}(\mathcal{A}).$$

It is monotone because

$$\mathcal{A} \subseteq \mathcal{B} \implies \text{cl}(\mathcal{A}) \subseteq \text{cl}(\mathcal{B}).$$

It is idempotent because the first closure already satisfies all heredity rules:

$$\text{cl}(\text{cl}(\mathcal{A})) = \text{cl}(\mathcal{A}).$$

The coefficient dictionary is finite, so repeatedly adding required parent terms terminates after finitely many additions. The resulting set is contained in every heredity-valid superset of $\mathcal{A}$, which proves minimality. $\square$

Let

$$\mathbf{m}^{\text{closed}} \in \{0, 1\}^{240} \tag{A.136}$$

denote the closed support mask.

A.20. Training-only selected-support refit

After candidate selection and heredity closure, coefficients outside the closed support are fixed to zero. The refit objective is

$$\mathcal{J}_{\text{refit}} = \mathcal{L}_{\text{data}} + \lambda_{\text{ridge}} \frac{\|\mathbf{m}^{\text{closed}} \odot \boldsymbol{\beta}^{\text{red}}\|_2^2}{240} + \lambda_{\text{phys}} \mathcal{B}_{\text{phys}}. \tag{A.137}$$

The sparse-group term is not used during refitting. Structural selection has already been completed, so the refit estimates the coefficients of the closed structure without continuing group shrinkage.

A.21. Calibration-stage candidate selection

For candidate job j , let $\mathcal{S}_{\text{cal}}$ denote the calibration trajectories and $\mathcal{K}_s^{\text{eval}}$ the valid evaluated samples of trajectory s . Define

$$N_s^{\text{eval}} = |\mathcal{K}_s^{\text{eval}}|. \tag{A.138}$$

The calibration trajectory RMSE is

$$\text{RMSE}_{j,s}^{\text{cal}} = \sqrt{\frac{1}{N_s^{\text{eval}}} \sum_{k \in \mathcal{K}_s^{\text{eval}}} e_{j,s,k}^2}. \quad (\text{A.139})$$

The trajectory-weighted calibration score is

$$E_j^{\text{cal}} = \frac{1}{|\mathcal{S}_{\text{cal}}|} \sum_{s \in \mathcal{S}_{\text{cal}}} \text{RMSE}_{j,s}^{\text{cal}}. \quad (\text{A.140})$$

Each calibration trajectory therefore receives equal weight.

Let

$$E_{\min}^{\text{cal}} = \min_j E_j^{\text{cal}}, \quad (\text{A.141})$$

and

$$\epsilon_{\text{cal,eq}} = 0.0025. \quad (\text{A.142})$$

The calibration-equivalent pool is

$$\mathcal{A}_{\text{cal}} = \{j : E_j^{\text{cal}} \leq (1 + \epsilon_{\text{cal,eq}}) E_{\min}^{\text{cal}}\}. \quad (\text{A.143})$$

Jobs in $\mathcal{A}_{\text{cal}}$ are ordered lexicographically by active group count, nonzero coefficient count, active candidate coordinate count, trajectory-weighted calibration RMSE, candidate identifier, regularization value, and job identifier. The first job is selected.

A.22. Model freezing and test isolation

Before ordinary-test or strict-test evaluation, the following objects are frozen:

$$\mathcal{F} = \{M^*, \lambda^*, \mathbf{m}^{\text{closed}}, \beta^*, \mathbf{q}_{\text{mid}}, \mathbf{q}_{\text{half}}, \mathbf{A}_{\text{id}}, \boldsymbol{\eta}^{\min}, \boldsymbol{\eta}^{\max}, \text{initial-state rule, simulation configuration}\}. \quad (\text{A.144})$$

Proposition A.3.. Under the implemented sequence, ordinary-test and strict-test observations cannot affect candidate identity, regularization selection, support selection, refitted coefficients, or scheduler normalization.

Proof. The training partition determines scheduler normalization, hierarchy weights, identifiable expansion, fitted candidate coefficients, and the selected-support refit. The calibration partition determines the candidate job and regularization choice. The frozen set $\mathcal{F}$ is completed before either test partition is opened. Ordinary-test and strict-test data are subsequently passed only to the fixed prediction and diagnostic functions. No component of $\mathcal{F}$ is recomputed after test access. $\square$

This is a procedural test-isolation result. It does not imply that the partitions are identically distributed, nor does it eliminate distribution shift.

A.23. Accuracy metrics

Let

$$\mathcal{E} = \{(s, k) : k \geq 1, m_{s,k}^{\text{eval}} = 1\} \quad (\text{A.145})$$

denote the pooled evaluation set. The pooled RMSE is

$$\text{RMSE}_{\text{pool}} = \sqrt{\frac{1}{|\mathcal{E}|} \sum_{(s,k) \in \mathcal{E}} e_{s,k}^2}. \quad (\text{A.146})$$

The pooled MAE is

$$\text{MAE}_{\text{pool}} = \frac{1}{|\mathcal{E}|} \sum_{(s,k) \in \mathcal{E}} |e_{s,k}|. \quad (\text{A.147})$$

The pooled bias is

$$\text{Bias}_{\text{pool}} = \frac{1}{|\mathcal{E}|} \sum_{(s,k) \in \mathcal{E}} e_{s,k}. \quad (\text{A.148})$$

For trajectory s ,

$$\text{RMSE}_s = \sqrt{\frac{1}{N_s^{\text{eval}}} \sum_{k \in \mathcal{K}_s^{\text{eval}}} e_{s,k}^2}. \quad (\text{A.149})$$

The trajectory-weighted RMSE is

$$\text{RMSE}_{\text{TW}} = \frac{1}{|\mathcal{S}_{\mathcal{E}}|} \sum_{s \in \mathcal{S}_{\mathcal{E}}} \text{RMSE}_s. \quad (\text{A.150})$$

The pooled RMSE weights every sample equally, whereas the trajectory-weighted RMSE weights every trajectory equally.

A.24. Residual-dependence diagnostics

For diagnostic variable $x_{s,k}$, the sample-weighted Pearson correlation is

$$\rho(e, x) = \frac{\sum_{(s,k) \in \mathcal{E}} (e_{s,k} - \bar{e}) (x_{s,k} - \bar{x})}{\sqrt{\sum_{(s,k) \in \mathcal{E}} (e_{s,k} - \bar{e})^2} \sqrt{\sum_{(s,k) \in \mathcal{E}} (x_{s,k} - \bar{x})^2}}. \quad (\text{A.151})$$

This statistic quantifies linear association. It does not establish general independence or causality.

For diagnostic bin $\mathcal{B}_b$,

$$\bar{e}_b = \frac{\sum_{(s,k) \in \mathcal{E}} \mathbf{1}\{x_{s,k} \in \mathcal{B}_b\} e_{s,k}}{\sum_{(s,k) \in \mathcal{E}} \mathbf{1}\{x_{s,k} \in \mathcal{B}_b\}}. \quad (\text{A.152})$$

For SOC bin $\mathcal{Z}_a$ and temperature bin $\mathcal{T}_b$,

$$n_{a,b} = \sum_{(s,k) \in \mathcal{E}} \mathbf{1}\{z_{s,k} \in \mathcal{Z}_a\} \mathbf{1}\{T_{s,k} \in \mathcal{T}_b\}, \quad (\text{A.153})$$

and

$$\bar{e}_{a,b} = \frac{\sum_{(s,k) \in \mathcal{E}} \mathbf{1}\{z_{s,k} \in \mathcal{Z}_a\} \mathbf{1}\{T_{s,k} \in \mathcal{T}_b\} e_{s,k}}{n_{a,b}}. \quad (\text{A.154})$$

The training scheduler-support count is

$$n_{a,b}^{\text{tr}} = \sum_{(s,k) \in \mathcal{E}_{\text{tr}}} \mathbf{1}\{\xi_{z,s,k} \in \Xi_a\} \mathbf{1}\{\xi_{T,s,k} \in \Theta_b\}. \quad (\text{A.155})$$

This describes sample density in the fitted SOC–temperature plane; it does not establish chemistry-, cell-, current-history-, or trajectory-level support.

A.25. Centered mini-batch gradient-span diagnostic

Let p_S be the number of selected reduced coefficients, and let

$$\mathbf{g}_b = \nabla_{\beta_S} \mathcal{L}_{\text{data}}^{(b)}(\beta^*) \in \mathbb{R}^{p_S} \quad (\text{A.156})$$

be the selected-support data-loss gradient for training mini-batch b . Stack B gradients into

$$\mathbf{G} = \begin{bmatrix} \mathbf{g}_1^\top \\ \vdots \\ \mathbf{g}_B^\top \end{bmatrix} \in \mathbb{R}^{B \times p_S}. \quad (\text{A.157})$$

The mean gradient is

$$\bar{\mathbf{g}} = \frac{1}{B} \sum_{b=1}^B \mathbf{g}_b, \quad (\text{A.158})$$

and the centered matrix is

$$\mathbf{G}_c = \mathbf{G} - \mathbf{1}\bar{\mathbf{g}}^\top. \quad (\text{A.159})$$

Equivalently,

$$\mathbf{G}_c = \mathbf{H}_B \mathbf{G}, \quad \mathbf{H}_B = \mathbf{I}_B - \frac{1}{B} \mathbf{1}\mathbf{1}^\top. \quad (\text{A.160})$$

Since

$$\text{rank}(\mathbf{H}_B) = B - 1, \quad (\text{A.161})$$

$$\text{rank}(\mathbf{G}_c) \leq \min(B - 1, p_S). \quad (\text{A.162})$$

The singular-value analysis follows standard numerical linear-algebra practice [11].

Let

$$\mathbf{G}_c = \mathbf{U}\Sigma\mathbf{V}^\top \quad (\text{A.163})$$

be the singular-value decomposition. The reported practical gradient-span rank is

$$r_{\text{grad}} = \#\{i : \sigma_i(\mathbf{G}_c) > \max(B, p_S)\epsilon_{\text{mach}}\sigma_1(\mathbf{G}_c)\}. \quad (\text{A.164})$$

This quantity measures the numerical span of centered mini-batch gradient directions at the selected solution. It is not the rank of the complete recursive residual Jacobian, not a Fisher-information rank, and not a full parameter-identifiability proof.

A.26. Summary of established properties

These properties establish the mathematical consistency of the model construction and validation sequence. They do not establish that the selected model is physically complete. Model adequacy must still be judged from held-out voltage error, strict-test behavior, residual dependence, chemistry- and cell-resolved diagnostics, and scheduler support.

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